AUTOMATED AND SCALABLE PLATFORM FOR SUBSCRIPTION MANAGEMENT WITH ALERTS

(Sub Track AI)

SUBMITTED BY

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Introduction

Subscription services are widely used today.

- Netflix
- Spotify
- Cloud Storage
- SaaS Tools

Users often subscribe to multiple services simultaneously.
From entertainment to productivity, the number of recurring payments is increasing.

> Managing these subscriptions manually becomes difficult.

Problem Overview

- Forgetting subscription renewal dates
- Unexpected recurring payments draining finances
- Paying for unused or "zombie" subscriptions
- No centralized platform to manage all subscriptions
- Poor visibility of total monthly expenses

Problem Statement

There is a critical need for a centralized system that addresses the fragmentation of modern subscription services:

Tracks All Subscriptions
A single dashboard to monitor active, paused, and cancelled services across all platforms.

Alerts Before Renewal
Automated notifications sent days before billing to prevent unwanted auto-renewals.

Provides Expense Analytics
Visual breakdowns of monthly and yearly spending habits categorized by service type.

Controls Recurring Spending
Empowers users to identify unused services and reduce unnecessary financial leakage.

Objectives of the Project

Centralized Management

Develop a unified platform to manage all subscription services in one place effectively.

Automated Alerts

Provide timely notifications to users before renewal dates to prevent unwanted charges.

Expense Tracking

Track monthly and yearly recurring expenses to give users better financial visibility.

Microservices Architecture

Implement a scalable microservices architecture to ensure system maintainability and flexibility.

Mobile Application

Develop a responsive mobile application using React Native for seamless user access.

Modern Tech Stack

Demonstrate the use of modern software technologies like Spring Boot, Docker, and JWT.

Scope of the Project

INCLUDED FEATURES

Add and manage subscriptions
Centralized hub for all services

Track renewal dates
Visual calendar and countdowns

View subscription dashboard
Analytics and expense overview

Receive alerts
Automated notifications before billing

NOT INCLUDED

Payment gateway integration
Direct payment processing is out of scope

Bank account integration
No auto-debit or bank API connections

AI-based recommendations
Advanced suggestion algorithms (future scope)

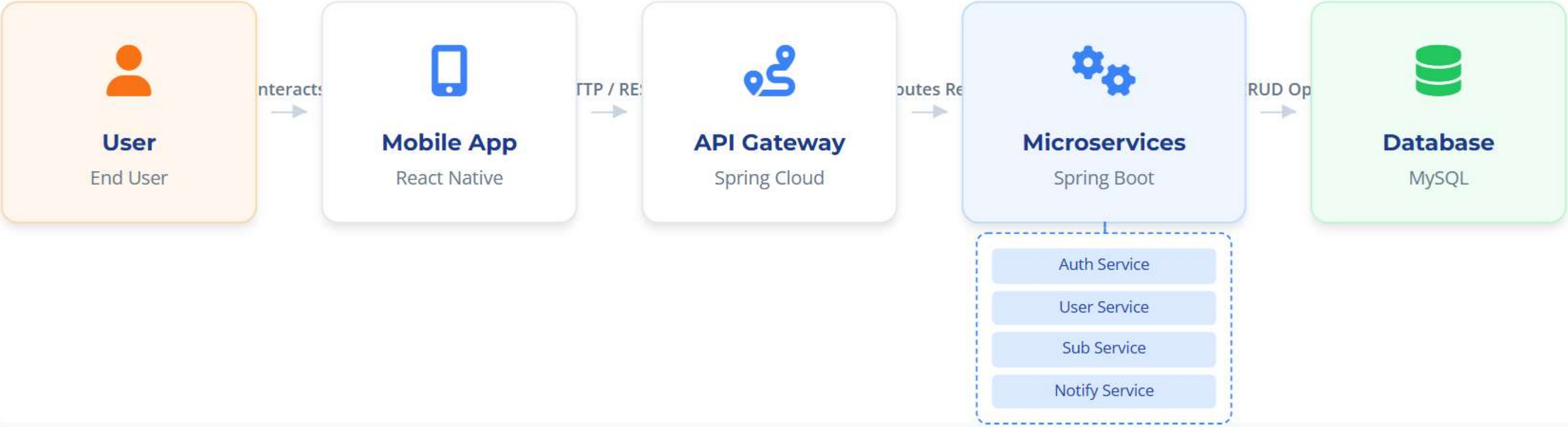
Technology Stack

LAYER	TECHNOLOGY
Backend	Java Spring Boot REST API
API Gateway	Spring Cloud Gateway Routing
Mobile App	React Native Cross-Platform
Database	MySQL Relational DB
DevOps	Docker Containerization

Tools Used

DEVELOPMENT TOOLS	PURPOSE
Spring Boot	Backend development
React Native	Mobile app development
MySQL	Database management
Docker	Containerized deployment
GitHub	Version control

System Architecture



Requests flow securely from the **Mobile App** through the **API Gateway**, which balances loads and routes traffic to the appropriate independent **Microservices**.

Microservices in System

MAIN SERVICES

Authentication Service
Handles login and JWT security

User Service
Manages user profiles and data

Subscription Service
Tracks renewals and plans

Notification Service
Sends alerts via email/push

KEY BENEFITS

Scalability
Scale individual services on demand

Maintainability
Easier to debug isolated components

Independent Deployment
Update services without system downtime

Work Completed Till Date

System Architecture Design
Defined microservices interactions & database schema

Backend Microservices Setup
Spring Boot services initialized & configured

Authentication Module
Secure login implemented with JWT tokens

Database Integration
MySQL setup & Subscription CRUD APIs active

Mobile Interface
Initial React Native screens developed

API Gateway
Spring Cloud Gateway routing configured

Overall Development Progress 75%

Milestones Achieved

Secure Authentication Implemented

JWT-based login and signup flow fully functional.

✓ DONE

Subscription Management Developed

CRUD operations for user subscriptions active.

✓ DONE

API Gateway Configured

Spring Cloud Gateway routing traffic to microservices.

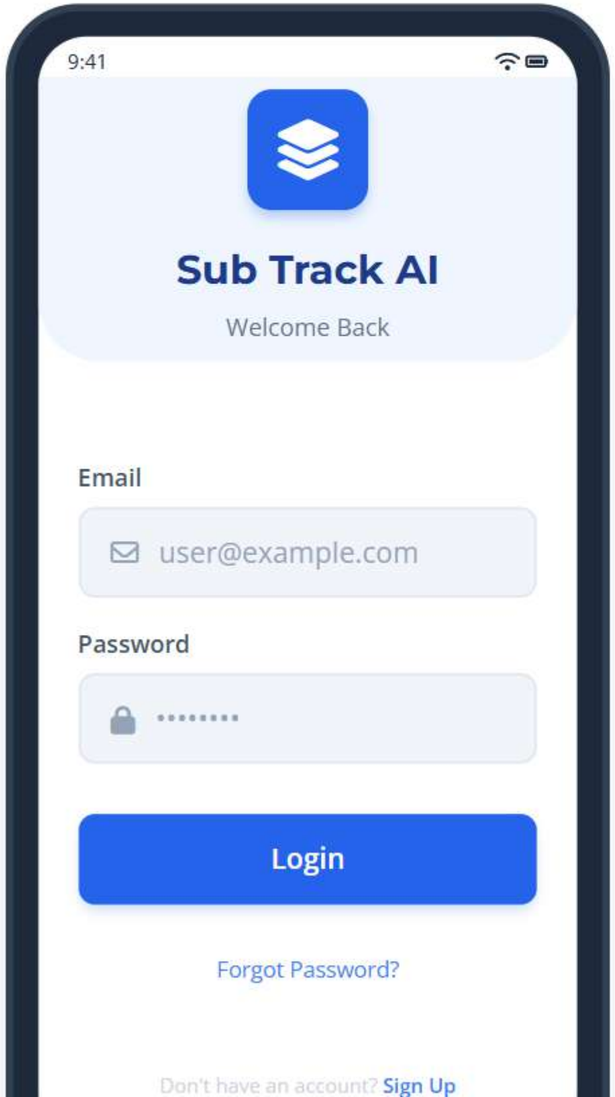
✓ DONE

Mobile App Prototype Created

React Native UI built for Login and Dashboard.

✓ DONE

Prototype (Login Screen)



Secure Authentication

Implements robust user verification. Passwords are never stored in plain text; they are hashed before storage in the database.

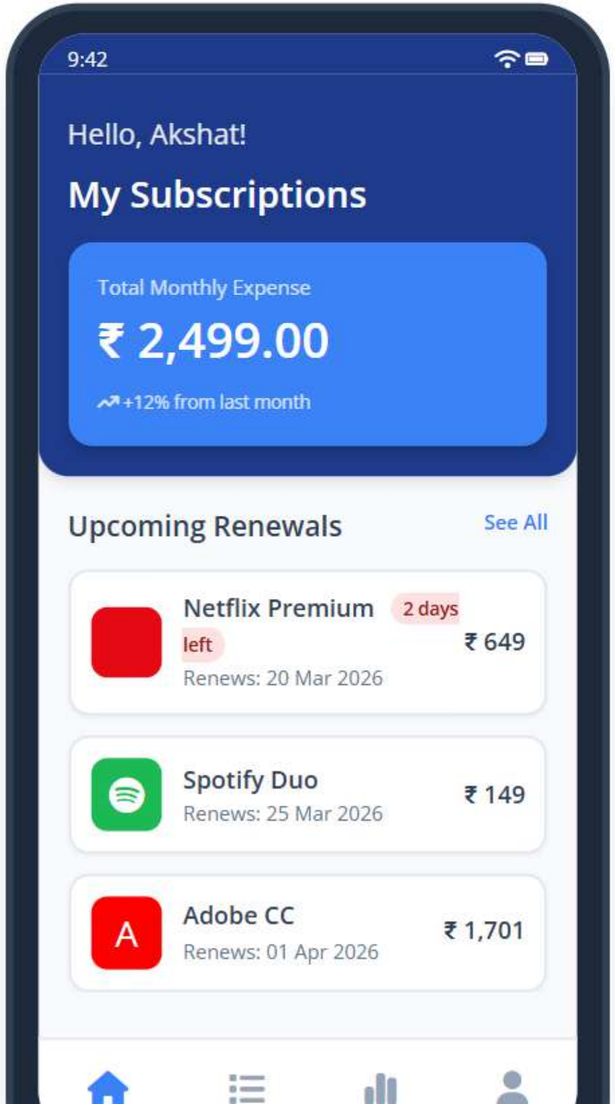
JWT Implementation

Upon successful login, a JSON Web Token (JWT) is issued. This token is used for subsequent API requests to microservices, ensuring stateless and secure sessions.

User Experience

Clean, minimal interface designed with React Native. Includes validation for email formats and password strength.

Prototype (Dashboard)



Expense Overview

Provides a clear summary of total monthly spending at a glance. Helps users understand their financial commitment instantly upon login.

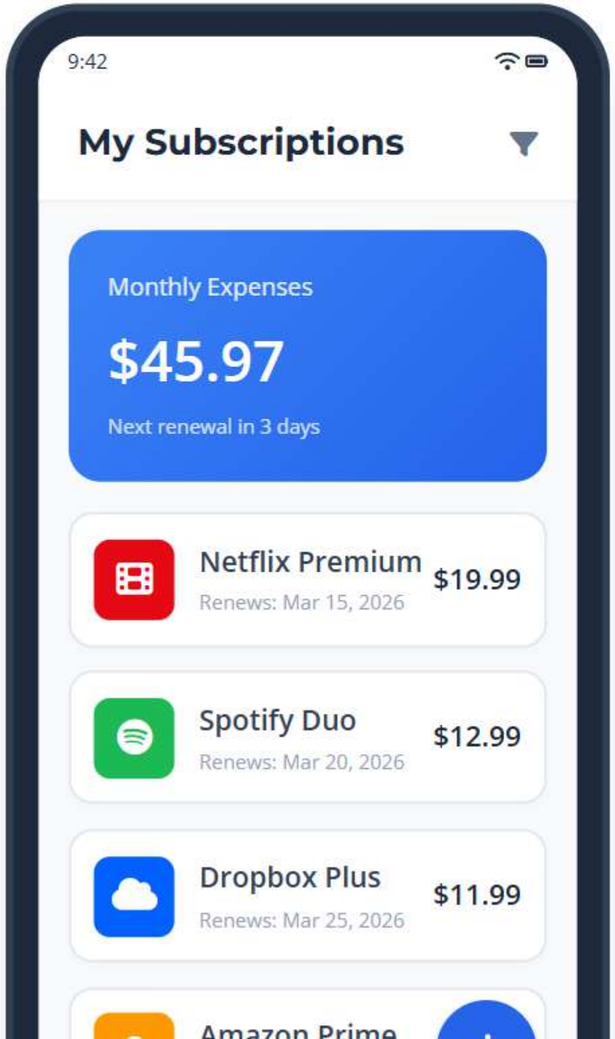
Upcoming Renewals

Highlights subscriptions that are due for renewal soon. Visual alerts (red badges) ensure critical dates are not missed.

Active Subscriptions

Lists all active services with key details like renewal date and cost. Users can tap to view more details or manage specific subscriptions.

Prototype (Subscription Page)



CRUD Operations

Complete management capability allowing users to Create, Read, Update, and Delete subscriptions. The list view provides instant access to edit any entry.

Billing Cycle Tracking

Visual indicators for renewal dates help users plan finances. The system calculates the next billing date automatically based on cycle settings.

Real-Time Sync

All changes are instantly synchronized with the MySQL database via the Subscription Microservice, ensuring data consistency across devices.

Challenges Faced

Developing a scalable microservices-based platform presented several key technical difficulties:

- Designing Microservices Architecture
- Secure Authentication Implementation
- Managing Service Communication
- Containerization with Docker

These challenges required careful planning of service boundaries, data consistency patterns, and secure token handling across the distributed system.

Solutions Implemented

Spring Cloud Gateway for Routing
Implemented a centralized API Gateway to handle request routing, load balancing, and standardized entry points for all microservices.

JWT Based Authentication
Adopted JSON Web Tokens for stateless, secure user verification, ensuring safe data access across distributed services without session storage.

REST API Communication
Established standardized RESTful endpoints to enable reliable and efficient data exchange between the frontend mobile app and backend services.

Docker for Containerized Deployment
Utilized Docker containers to package microservices with their dependencies,

Future Work

COMMUNICATION

Automated Notification System

Implementation of a background processor for timely renewal alerts and integration of Push Notifications for real-time user updates.

DATA INTELLIGENCE

Advanced Analytics Dashboard

Deep-dive financial insights including expense trend analysis, category-wise breakdown, and spending forecasts to help users budget better.

USER EXPERIENCE

Improved Mobile Interface

Enhancing the React Native UI with smoother animations, dark mode support, and more intuitive subscription management workflows.

Expected Outcomes

Centralized Tracking

A unified dashboard for monitoring all active subscriptions and recurring payments in one place.

Automated Alerts

Timely notifications before renewals to prevent unwanted charges and ensure user control.

Financial Awareness

Improved visibility into spending habits with detailed monthly and yearly expense analytics.

Microservices Architecture

Successful implementation of a decoupled, service-oriented backend using Spring Boot.

Scalable & Efficient System

A robust, containerized platform capable of handling growing user data and high traffic loads efficiently.

Thank You!

For your time and attention

Any Questions?

We are happy to answer any queries regarding the project.